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RESEARCH ARTICLE

Forecasting number of ISO 14001 certifications of selected countries: application of even GM (1,1), DGM, and NDGM models

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Abstract
The adaptability of ISO 14001 is considered as one of the most useful tools for environmental sustainability and worldwide competitive advantage; however, the future of ISO 14001 certification faces some uncertainties because of its uneven acceptance in various countries. These uncertainties, if not properly managed, can hinder the implementation of business management systems in these countries. In order to guide policymakers in better management of ISO 14001 in future with certainty, this study aims to forecast the ISO 14001 certifications for 10 years for China, India, the USA, Italy, Japan, and Germany, the top six certified countries, through advanced mathematical modeling, namely grey models, even GM (1,1), discrete GM (1,1), and non-homogenous discrete grey model (NDGM). The benefits of mentioned models are ensured accuracy in assessment using small samples and poor information. Moreover, current research is a pioneer in the certifications growth analysis using the Synthetic Relative Growth Rate and Synthetic Doubling Time models. Finally, the empirical analysis indicated that China is constantly leading in terms of its ISO 14001 certifications till 2026 and the performance of developing countries was spectacular. Furthermore, the article has proposed some suggestions for the policymakers to make the environment more sustainable.

Keywords ISO 14001 · Forecasting · Environmental sustainability · Competitive advantage · Doubling time · Relative growth rate

Introduction

Since 1996, the ISO 14000 series certified organizations have continuously demonstrated the great reorganization and wide popularity of the certification due to the growing concerns about climate change. In 2015, the Conference of Parties (COP) of the United Nations Framework Convention for Climate Change (UNFCCC) agreed to limit the increase in the global temperature to 2 °C above pre-industrial levels by 2020 (United Nations Framework Convention on Climate Change 2016). After that, Intergovernmental Panel on Climate Change (IPCC) emphasized that limiting global warming to 1.5 °C will be a challenging task in the years ahead (Science Insight 2019). Hence, the future forecasting by using the most accurate and well-accepted methodologies can guide the specific measurement criteria at the micro and macro level. As we know, since the pre-industrial era, a gradual increase in emission of greenhouse gases (GHG) has exceedingly affected the human and natural system due to economic and population growth, and it is now higher than ever (Change 2014). Continued increase of industrial pollution in the environment causes further global warming and long-lasting effects on all components of the ecosystem. Many models and techniques have been developed to mitigate the emissions and pollutants yet. Environmental Management System (EMS) ISO 14001 is one of them and was published in the mid-1900s by the International Organization for Standardization (ISO). Environmental sustainability is a big challenge for companies nowadays because International Energy Outlook (IEA 2015) reports that CO_2 emissions due to combustion of fossil fuels are worsening, and this is directly

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associated with the climate change context. The report predicts that worldwide energy-induced CO_2 emissions will increase by 35.6 billion metric tons in 2020, increasing further by 7.6% in 2040 to reach a value of 43.2 billion metric tons.

Several studies have been conducted to understand the importance of ISO 14001 and its contribution toward environmental performance (Campos et al. 2015; Neves et al. 2017). ISO 14001 introduces companies to their environmental stewardship to draw broad interest toward their customers worldwide (Darnall and Carmin 2005). From improved efficiencies, it offers cost-effective benefits to companies, and from a social perspective, ISO 14001 is a roadmap to increase the quality of life by reducing the environmental and regional risks such as health problems, food insecurity, heat waves, and floods. The implementation of ISO 14001 provides some external and internal benefits to organizations, as it externally increases customer confidence, improves public image and community relationship, provides internal improvement in business processes, reduces liability and incidents, and enhances efficiency and cost-saving (Strachan et al. 2003). Numerous studies have been widely conducted on ISO 14001 in different sectors in the last decades (Vastag et al. 2004; Zeng et al. 2005; Prakash and Potoski 2006; Uchida and Ferraro 2007; Gavronski et al. 2008; Cañón-de-Francia and Garcés-Ayerbe 2009; Wahba 2010; Qi et al. 2011; Mohee et al. 2012; Khan 2013; Singh et al. 2014; Ismail et al. 2014; Daddi et al. 2015; Jabbour 2015; Ferrón-Vílchez 2016; Hojnik and Ruzzier 2017; Salim et al. 2018; Graafland 2018). Also, in the literature (Llach et al. 2011, 2015; Sampaio et al. 2011; To and Lee 2014), one finds ISO 9001 forecasting model in association with a statistical technique such as regression analysis for the prediction of the future of Quality Management System between the period 2007 to 2010.

The implementation of EMS mitigates the weaknesses of traditional regulations and provides a suitable alternative for companies to gain sustainability, reduction in high cost and improvement in the corporate image (McGuire 2014). Similarly, Arimura et al. (2016) also illustrated that the adaptability of EMS leads to minimization of companies' CO_2 emission level and causes a reduction in the overall pollution. There is a direct relationship between a company's competitive position and ISO 14001 implementation, as noted by Giménez Leal et al. 2003. For more clarification, the relationship and differences between ISO 14000 and 14001 standards ought to be addressed. In fact, ISO 14000 is a global standard that supplies a framework for all firms that want to enhance their environmental management efforts. The ISO 14000 standard consists of several standards while ISO 14001 is one of the most important among them. It should be noted that ISO 14001 determines the necessities of an EMS for all size of organizations. Also, the ISO 14001 standard encourages the plan-check-do-review-improve cycle. The objective of the current study is the empirical prediction of ISO 14001 certifications for the year 2017–2026 for the six selected countries China, Japan, India, Germany, Italy, and the USA, based on the data available at ISO (2018) from 2005 to 2016 while executing advanced mathematical modeling, namely grey forecasting models, i.e., even grey model GM (1,1), discrete GM (1,1), and non-homogenous discrete grey model (NDGM). Synthetic relative growth rate (RGR) and synthetic doubling time (D_t), as proposed by Javed and Liu (2018), were used to analyze the certification growth and comparison among these countries. The grey forecasting models, especially the GM (1,1) models, can be superior to other forecasting models in the context of small samples and poor information (Zhou and He 2013). Hence, current research has utilized grey models for forecasting. The study would increase the understanding of possible expected ISO 14001 performance in China, Japan, India, the USA, Italy, and Germany in the upcoming years. Finally, the study investigated the future trends in the ISO 14001 performance so that policy makers can make decisions using the future expected outcomes. The study provides valuable information for decision makers in a holistic way.

This paper is organized as follows: Section 1 is the introduction; Section 2 presents the background/literature review of Environmental Management System (EMS) ISO 14001 and the current scenario in developed and developing countries; Section 3 describes the method we used to forecast ISO 14001 in the given scenario; Section 4 presents the results. Finally, Section 5 concludes the study, with limitations and future research direction.

Literature review

In 1987, a report was published by Commission on the Environment and Development of the United Nations with future agenda, in which the definition of sustainable development is to meet the present and future generations' needs without compromising their abilities (Brundtland et al. 1987). This report inaugurated new routes for a decision support system by effective environmental management. ISO and International Electrotechnical Commission (IEC) became partners in development and environmental standard (ISO 2012a) in Rio de Janeiro when the United Nations hosted the earth summit in 1992. A technical committee was formed by ISO and IEC to inspect the development of environmental management system a few months after their partnership. In 1996, Environment Management System was launched by ISO and has been updated twice, in 2004 and 2015. According to ISO (2012b), the ISO 14001:2015 has the following properties: (a) setting and systematically achieving environmental goals and objectives and illustrating the goals and objectives achieved; (b) internal and external scanning for identification of relevant environmental aspects to control the companies activities, products, and services influenced by environmental impact; (c). ISO 14001 is a tool used by companies to continuously observe environmental performance. ISO 14001 is a general environmental standard for all types of organizations, the main objective of the standard is to construct a framework for organizations to set up their goals, policies, and actions. The adoption ratio of EMS certification varies from country to country. Prakash and Potoski (2006) analyzed the actual data of 108 countries and investigated whether adoption rate was higher in those countries where trading partners adopted this innovative and friendly environmental mechanism for consumer attraction. The lack of EMS diffusion studies and a logistic curve were used to analyze the diffusion of ISO 14001 certification with a dataset of 13 countries from 1995 to 2005. Casadesús et al. (2008) reported in their study that in 2005 ISO 14001 certification saturation level was 64.6%, with a maximum number of certification it would be 160,000. Casadesús et al. (2008) also argued that the saturation would move to 95% in 2008 in most of the 13 selected countries, but later, unfortunately, the situation was different at the global and country level. Qi et al. (2011) reported in their study diffusion of ISO 14001 effected positively in China because of community stakeholder and foreign customer demand. Likewise, Marimon et al. (2011) investigated the diffusion of ISO 14001 and argued that diffusion among sectors on global and specific country level tend to be undiversified. Furthermore, the diffusion of EMS was analyzed by using logistic curve function technique and by the end of 2010 a 51% increase in certification and tendency toward saturation at 379900 worldwide were illustrated. They also argued that there would be a 50,000 increase in ISO 14001 certification in 2011 and 2012, and 45,800 from the start of 2013 to the end of 2015. The Far East followed by Europe, and Africa/West Asia will be the most prominent in some new ISO 14001 certifications. Castka and Prajogo (2013) have studied the relationship between the pressure of secondary stakeholders and ISO 14001. In order to investigate the issue, they have chosen 328 companies from Australia and New Zealand. Based on the results, ISO 14001 is an acceptable certificate from secondary stakeholders, yet pressure from secondary stakeholders has not affected internalization of ISO 14001 significantly. Testa et al. (2014) believed that the impact of ISO 14001 and the EMAS should be addressed to enhance environmental performance. Hence, they have investigated the effect of two mentioned standards on 229 energy plants in Italy. They found that ISO 14001 and EMAS have a significant effect on environmental performance. Nguyen and Hens (2015) have investigated the influence of ISO 14001 in Vietnam's cement plants. In their research, they have tried to use a questionnaire for comparing certified and non-certified cement companies. Their results illustrated that ISO 14001 is a potential tool in order to promote environmental performance in developing nations with increasing growth ratio. Boiral et al. (2018) have reviewed the papers that have been published between 1996 and 2015. They illustrated that most researches have presented optimistic results and the influence of ISO 14001 on environmental performance is shown. Their results have provided a suitable framework for managers in order to make a desirable decision. Ozusaglam et al. (2018) believed that EMS is the most important tool for management to reduce the impact of a firm's activities on the environment. Based on the results, implementation of EMS can lead to win-win opportunities for companies and early adopters can achieve European market. Iatridis and Kesidou (2018) have investigated the role of internal motivation and external pressure on the implementation of ISO 14001. They believe that internal motivation plays a vital role in the implementation of ISO 14001 and external pressures have an effect on internalization of ISO 14001, but their relationship may not be the linear type.

Current scenario of ISO 14001

In the recent decade, the popularity of ISO 14001 standard has been increased worldwide (Prajogo et al. 2014), but still with a comparison of developing and developed countries we can see the uneven adoption of ISO 14001 (Neumayer and Perkins 2004). According to ISO (2018), in Asia, the number of certifications increased from 5234 to 173,324 during 2000–2015, meanwhile European countries have moved from 7235 certifications in 2000 to 119,754 in 2015. Although in the USA the growth of ISO 14001 certifications is not like other developed countries, every year, many American organizations apply for ISO 14001 certification to adopt green processes (Lee et al. 2017). In the current scenario, the significance of EMS has increased to justify the fact that the environmental performance of organizations has improved by controlling their production processes and services on the environment (Rino and Salvador 2017). Despite that, the growth of ISO 14001 certification in Asia is increasing such as China, India, Japan, and South Korea. Neumayer and Perkins (2004) suggested that the low adoption of ISO 14001standard in developed and developing countries will result in the exclusion of uncertified companies, which in return could serve those countries where companies are not ISO 14001 certified. Morrow and Rondinelli (2002) reported that EMS and other international standard registration is gradually increasing by organizations around the world. This scenario may impose pressure on companies to adopt ISO 14001 certification for environmental sustainable development (Prajogo et al. 2012). Likewise the rising trend of ISO 14001 adoption gives a competitive advantage to the organizations in the international market (Yu and Choi 2016). In developed regions like the USA and Europe, adoption of ISO 14001 (Morrow and Rondinelli 2002) is an indicator of environmental sustainability and a tool for competitive advantage.

Methodology

In this section, the models used have been discussed, along with the data collection methods.

Data collection and analyses

In the current study, environmental sustainability indicates ISO 14001 certification. First of all, top ISO 14001 accreditated countries globally were selected. They were China, Japan, Italy, Germany, India, and the USA. The data used in this paper is from 2005 to 2016, collected from the International Organization for Standardization (ISO)'s official website (ISO 2018). Nanjing University of Aeronautics and Astronautics (NUAA)'s grey system software (v8.0) was used to forecast ISO 14001 certifications output, from the period 2017–2026 by even grey model GM (1,1), discrete grey model GM (1,1), and non-homogenous discrete grey model (NDGM). However, NDGM (1,1) was solved with the help of MATLAB and MS Excel. This data analysis and modeling methodology have been previously successfully executed by Javed and Liu (2018).

Grey forecasting models

Even grey model GM (1,1)

The concept of a grey system for the first time was proposed by Deng Julong in 1982 as a scientific theory for forecasting uncertain system involving small and poor data (Zhao and Guo 2016; Javed and Liu 2018; Javed et al. 2018). In many cases, grey system theory–based predictions have outperformed the predictions made through statistics (Javed and Liu 2018; Javed et al. 2018). There are different types of grey models in order to forecast data, some of which current research will examine. Fractional order accumulation is a new model of the grey system that can emphasize recent data instead of considering old and new data with the same weight (Wu et al. 2013). GM (1,1) is a basic and key component of grey system theory that is used for a prediction using limited data, or small sample size. The applications of grey system forecasting model GM (1,1) have been used in numerous studies such as rural household net income (Zhao et al. 2012), wind speed (El-Fouly et al. 2006), industrial output (Wang and Hsu 2008), electricity demand in China (Zhou et al. 2006), and research publications' analysis (Javed and Liu 2018). GM (1,1) presents one variable with first-order grey model to obtain forecasting information with the original sequence of data. There are four basic kinds of GM (1,1), even grey model is one of the most reliable models with non-exponential increasing sequence proved by (Liu et al. 2016), as follows in Eq. 1:

$$\hat{x}^{(0)}(k) = (1-\mathrm{e}^{a})\left(x^{(0)}(1)-\frac{b}{a}\right)\mathrm{e}^{-a(k-1)}, \quad k = 1,2,\ldots n \quad (1)$$

$x^{(0)}$ represents the original sequence of actual data in the above equation, further parameters a represents development index and reflects the trend of $\hat{x}^{(0)}$ and $\hat{x}^{(1)}$ while b is called grey actuating quantity. These are the contacts of an even GM (1,1) model. As literature reveals, the predictions are most suitable in data sequence for the first two data values after the last origin value.

$$\hat{x}^{(1)}(k) = \left(x^{(0)}(1)-\frac{b}{a}\right)e^{-a(k-1)} + \frac{b}{a}, \; k = 1,2,\ldots n \qquad (2)$$

Discrete grey model GM (1,1)

Even though results obtained from EGM are reliable and satisfactory, more improved outcomes are needed, thus we used DGM. The accuracy of GM (1,1) is an important issue; many scholars pay attention to solving this issue to obtain more accurate results and remove errors. Xie and Liu (2009) proposed a study in which they investigated the relationship between even GM and DGM with the help of McLaurin formula. They proved that the accuracy level of DGM and its optimization model is better than GM (1,1). In our case, the MAPE (%) of DGM turned out to be significantly better than even GM (1,1), as indicated in Table 7 of the current paper. If $\mathrm{x}^{(0)}$ represents the actual data sequence and $\mathrm{x}^{(1)}$ indicates the general accumulated sequences, then the discrete grey model is followed by (Liu et al. 2016):

$$\check{x}^{(0)}(k) = \left[x^{(0)}(1)-\frac{\beta 2}{1-\beta 1}\right]\beta 1^{k} + \frac{\beta 2}{1-\beta 1} \qquad (3)$$

Non-homogenous discrete grey model

The mechanism of NDGM is followed by the law of approximation non-homogenous exponential growth (Wu et al. 2014) based on the assumption of a sequence of actual/original data. Xie et al. (2013) suggested that the original data sequence is an agreement with a homogenous trend like GM (1,1). They also indicated that NDGM accuracy level is much better than DGM (1,1) and GM (1,1) in terms of mean sequence value and value set of the interval (Xie and Liu 2015). In our case, Table 7 indicated the overall accuracy level of NDGM is 95.47% among the six countries that was higher than DGM and EGM. NDGM has been applied in different fields such as Pirthee (2017) that predicted tourism in Mauritius and investigated whether NDGM forecasting accuracy level is better than another grey model. Meanwhile, Duan et al. (2017) forecasted the traffic flow in China and illustrated NDGM showed better performance than other grey forecasting models.

If the same sequence follows the non-homogenous discrete grey model $x^{(0)}$ and $x^{(1)}$ as above in GM (1,1), DGM (1,1) in Eqs. 1, 2, and 3, then:

$$\hat{x}^{(1)}(k+1) = \beta_1 \hat{x}^{(1)}(k) + \beta_2 .k + \beta_3 \tag{4}$$

$$\hat{x}^{(1)}(1) = x^{(1)}(1) + \beta_4 \tag{5}$$

$\hat{x}^{(1)}$ (k) is the simulated value of $x^{(1)}$ with four parameters β_1, β_2, β_3, and, β_4. We can rewrite the equation in matrix form if $k = 1, 2, 3, \ldots .n{-}1$

$$\begin{bmatrix} x^{(1)}(2) \\ x^{(1)}(3) \\ \vdots \\ x^{(1)}(n) \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \begin{bmatrix} x^{(1)}(1) & 1 & 1 \\ x^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ x^{(1)}(n-1) & n-1 & 1 \end{bmatrix}$$

For input data, the following matrix equation satisfies parameters β1, β2, β3, and, β4, which are the constants with a fixed value in each case, by applying the following relation in Eq. 6:

$$\hat{\beta} = \left(B^{\mathrm{T}}B\right)^{-1} B^{\mathrm{T}} Y = (\beta_1, \beta_2, \beta_3)^{\mathrm{T}} \tag{6}$$

Where

$$B = \begin{bmatrix} x^{(1)}(1) & 1 & 1 \\ x^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ x^{(1)}(n-1) & N-1 & 1 \end{bmatrix}, Y = \begin{bmatrix} x^{(1)}(2) \\ x^{(1)}(3) \\ \vdots \\ x^{(1)}(n) \end{bmatrix},$$

By using the following formula to calculate β4 for minimizing the sum of square error:

$$\beta_4 = \frac{\sum\limits_{k=1}^{n-1} \left[x^{(1)}(k+1) - \beta_1^k x^{(1)}(1) - \beta_2 \sum\limits_{j=1}^{k} j\beta_1^{k-j} - \frac{1-\beta_1^k}{1-\beta_1} .\beta_3 \right] .\beta_1^k}{1 - \sum\limits_{k=1}^{n-1} \left(\beta_1^k\right)^2} \tag{7}$$

$$\hat{x}^{(1)}(k+1) = \beta_1^k \hat{x}^{(1)}(1) + \beta_2 \sum_{j=1}^{k} j\beta_1^{k-j} + \frac{1-\beta_1^k}{1-\beta_1} \cdot \beta_3; \quad k = 1, 2, \ldots, n-1 \tag{8}$$

Liu and Forrest (2010) can be referred for further information about NDGM, its properties, and its parameters. This model can easily be executed on Matlab software.

Approach for comparative analysis of the performance of the models

`Mean absolute percentage error (MAPE) highlights the performance criteria for comparative analysis between the three grey models.

$$\mathrm{MAPE}(\%) = \frac{1}{n} \sum_{k=1}^{n} \left| \frac{x^{(0)}(k) - \hat{x}^{(0)}(k)}{x^{(0)}(k)} \right| \times 100\% \tag{9}$$

Here, $x^{(0)}(k), \hat{x}^{(0)}(k)$ represent forecasting data values (Javed and Liu 2018).

Certification growth and doubling time analysis

To the best of our knowledge, there is no certificate growth analysis model available. Therefore, publication growth analysis model was used as an alternative considering the similarities between the number of certifications and number of publications. For this purpose, literature suggests using relative growth rate, which is taken as ISO 14001 certificates' relative growth rate, and doubling time models. The doubling time (D_t) and relative growth rate (RGR) were calculated by following (Javed and Liu 2018). They applied two parameters (D_t and RGR) to predict certification growth of four countries by using EGM and NDGM. The formula of RGR is given by,

$$\mathrm{RGR} = (\ln N_2 - \ln N_1)/(t_2 - t_1) \tag{10}$$

Where N_2 and N_1 represent the cumulative number of certifications in a respective year t_2 and t_1. In our case, the above equation can be rewritten because $t_2 - t_1$ is 1 year.

$$\mathrm{RGR} = \ln (N_2/N_1) \tag{11}$$

Whereas doubling time (D_t) indicates time needed to become ISO 14001 double certification in terms of numbers in specified RGR. The (D_t) is given by:

$$D_t = (t_2 - t_1) \ln [2/(\ln N_2 - \ln N_1)] \tag{12}$$

Meanwhile, we can reduce this equation to this state:

$$D_t = \ln [2/(\ln N_2 - \ln N_1)] \tag{13}$$

However, if the *RGR* and D_t produce a different trend in the future as compared to that of a past trend then Javed and Liu (2018) solved this issue by proposing synthetic relative growth rate ($\mathrm{RGR}_{\mathrm{synthetic}}$) and synthetic doubling time ($D_{\mathrm{synthetic}}$) models. They defined the synthetic relative growth rate ($\mathrm{RGR}_{\mathrm{synthetic}}$) model in Eq. (14):

$$\mathrm{RGR}_{\mathrm{synthetic}} = \theta\left(\mathrm{RGR}_{\mathrm{past}}\right) + (1-\theta)\left(\mathrm{RGR}_{\mathrm{future}}\right) \tag{14}$$

Where $\mathrm{RGR}_{\mathrm{past}}$ indicates the relative growth rate obtained from original data and $\mathrm{RGR}_{\mathrm{future}}$ indicates the relative growth rate obtained for forecasted data. Here, θ indicates the coefficient of relative weights and value of θ can be taken as 0.5, generally.

Table 1 ISO 140001 certification growth for China

Years	Actual Data	EGM	DGM	NDGM	Cumulative	RGR	Mean RGR	Dt	Mean Dt
2005	12,683	12,683	12,683	12,683	12,683	–			
2006	18,842	28,436	28,582	25,185	31,525	0.911	0.374	0.787	1.825
2007	30,489	33,253	33,425	31,077	62,014	0.677		1.084	
2008	39,195	38,886	39,088	37,725	101,209	0.490		1.407	
2009	55,316	45,473	45,710	45,224	156,525	0.436		1.523	
2010	58,116	53,176	53,455	53,684	214,641	0.316		1.846	
2011	63,460	62,184	62,511	63,229	278,101	0.259		2.044	
2012	67,874	72,717	73,102	73,996	345,975	0.218		2.215	
2013	80,292	85,035	85,487	86,144	426,267	0.209		2.260	
2014	98,979	99,440	99,971	99,849	525,246	0.209		2.260	
2015	114,303	116,284	116,909	115,311	639,549	0.197		2.318	
2016	137,230	135,982	136,716	132,753	776,779	0.194		2.331	
2017		159,017	159,879	152,432	152,432				
2018		185,953	186,967	174,633	327,065	0.763	0.329	0.963	1.916
2019		217,453	218,644	199,679	526,744	0.477		1.434	
2020		254,288	255,688	227,935	754,679	0.360		1.716	
2021		297,363	299,007	259,812	1,014,491	0.296		1.911	
2022		347,735	349,667	295,775	1,310,266	0.256		2.056	
2023		406,640	408,909	336,348	1,646,614	0.228		2.169	
2024		475,522	478,189	382,120	2,028,734	0.209		2.260	
2025		556,073	559,206	433,758	2,462,492	0.194		2.334	
2026		650,269	653,949	492,015	2,954,507	0.182		2.396	
MAPE (%)		9.57	8.29	7.9					

The synthetic doubling time ($D_{synthetic}$) model is given in Eq. (15):

$$D_{synthetic} = \theta\left(D_{past}\right) + (1-\theta)\left(D_{future}\right) \tag{15}$$

zWhere, D_{past} indicates the doubling time got from actual data and RGR_{future} reports the relative growth rate got from simulated/forecasted data, and θ indicates the relative weight.

Results and discussion

In this study, we used three grey models—EGM, DGM, and NDGM—to forecast the ISO 14001 certification output among six countries. By using given data set from 2005 to 2016 by applying even grey model GM (1,1), discrete grey model GM (1,1), and non-homogenous discrete grey model NDGM (1,1), the calculated simulated

Table 2 ISO 140001 certification growth for Japan

Years	Actual data	EGM	DGM	NDGM	Cumulative	RGR	Mean RGR	Dt	Mean Dt
2005	23,466	23,466	23,466	23,466	23,466	–			
2006	22,593	23,466	31,766	31,696	46,059	0.674	0.244	1.087	2.381
2007	27,955	31,564	31,190	30,991	74,014	0.474		1.439	
2008	35,573	31,035	30,626	30,378	109,587	0.392		1.628	
2009	39,556	30,514	30,071	29,845	149,143	0.308		1.870	
2010	34,852	30,002	29,527	29,383	183,995	0.210		2.254	
2011	30,397	29,499	28,992	28,981	214,392	0.153		2.571	
2012	27,774	29,004	28,467	28,632	242,166	0.122		2.798	
2013	23,723	28,518	27,952	28,329	265,889	0.093		3.063	
2014	23,581	28,040	27,446	28,066	289,470	0.085		3.159	
2015	26,069	27,569	26,949	27,837	315,539	0.086		3.144	
2016	27,372	27,107	26,461	27,639	342,911	0.083		3.180	
2017		26,652	25,982	27,466	27,466				
2018		26,205	25,511	27,316	54,782	0.690	0.257	1.064	2.257
2019		25,766	25,049	27,186	81,969	0.403		1.602	
2020		25,334	24,596	27,073	109,042	0.285		1.947	
2021		24,909	24,150	26,975	136,017	0.221		2.203	
2022		24,491	23,713	26,890	162,906	0.180		2.406	
2023		24,080	23,284	26,816	189,722	0.152		2.574	
2024		23,676	22,862	26,751	216,473	0.132		2.719	
2025		23,279	22,448	26,695	243,169	0.116		2.845	
2026		22,889	22,042	26,647	269,816	0.104		2.957	
MAPE (%)		9.57	8.29	7.9					

Table 3 ISO 14001 certification growth for Italy

Years	Actual data	EGM	DGM	NDGM	cumulative	RGR	Mean RGR	Dt	Mean Dt
2005	7080	7080	7080	7080	7080	–			
2006	9825	11,322	11,342	10,002	16,905	0.870	0.305	0.832	2.059
2007	12,057	12,325	12,345	11,650	28,962	0.538		1.312	
2008	12,922	13,417	13,437	13,275	41,884	0.369		1.690	
2009	14,542	14,605	14,626	14,875	56,426	0.298		1.904	
2010	17,675	15,899	15,920	16,452	74,101	0.272		1.993	
2011	17,340	17,308	17,329	18,005	91,441	0.210		2.253	
2012	19,512	18,841	18,862	19,536	110,953	0.193		2.336	
2013	21,300	20,510	20,531	21,043	132,253	0.176		2.433	
2014	22,616	22,327	22,347	22,529	154,869	0.158		2.539	
2015	22,350	24,305	24,324	23,993	177,219	0.135		2.697	
2016	26,655	26,458	26,476	25,435	203,874	0.140		2.658	
2017		28,802	28,818	26,856	26,856				
2018		31,353	31,368	28,255	55,111	0.719	0.278	1.023	2.136
2019		34,131	34,143	29,634	84,746	0.430		1.536	
2020		37,154	37,164	30,993	115,739	0.312		1.859	
2021		40,446	40,452	32,332	148,071	0.246		2.094	
2022		44,029	44,030	33,651	181,722	0.205		2.279	
2023		47,929	47,926	34,950	216,672	0.176		2.431	
2024		52,175	52,166	36,230	252,902	0.155		2.560	
2025		56,797	56,781	37,492	290,394	0.138		2.672	
2026		61,829	61,805	38,734	329,129	0.125		2.771	
MAPE (%)		4.53	4.40	3.14					

values are presented in Tables 1, 2, 3, 4, 5, and 6. Interesting results are obtained; Table 1 represents the results for China. The values calculated by MAPE % indicate the accuracy level of EGM, DGM, and NDGM, acquired 90.43%, 91.71%, and 92.1% respectively, which implies that NDGM is best fit grey model and showed a slightly more effective method to predict ISO 14001 certification output for six countries as compared to EGM and DGM. According to EGM parameters $a = -0.15648621$ and $b = 24{,}284.5233$, Eq. (16) will be obtained:

$$x^0(N) = 407928.4799\left(1 - e^{-0.15648621}\right)e^{0.15648621(N-1)} \quad (16)$$

These a and b parameters can be calculated by the software mentioned in the "Methodology" section.

Table 4 ISO 14001 certification growth for Germany

Years	Actual data	EGM	DGM	NDGM	Cumulative	RGR	Mean RGR	Dt	Mean Dt
2005	4440	4440	4440	4440	4440	–			
2006	5415	4873	4879	5071	9855	0.797	0.306	0.920	2.211
2007	4877	5183	5189	5292	14,732	0.402		1.604	
2008	5709	5513	5518	5544	20,441	0.328		1.809	
2009	5865	5864	5869	5829	26,306	0.252		2.070	
2010	6001	6238	6242	6152	32,307	0.205		2.276	
2011	6254	6635	6639	6519	38,561	0.177		2.425	
2012	7015	7057	7061	6935	45,576	0.167		2.482	
2013	7983	7506	7509	7406	53,559	0.161		2.517	
2014	7702	7984	7987	7941	61,261	0.134		2.700	
2015	8224	8493	8494	8547	69,485	0.126		2.765	
2016	9444	9033	9034	9234	78,929	0.127		2.753	
2017		9608	9608	10,013	10,013				
2018		10,220	10,219	10,897	20,910	0.736	0.307	0.999	2.000
2019		10,871	10,868	11,899	32,808	0.450		1.491	
2020		11,563	11,559	13,035	45,843	0.335		1.788	
2021		12,299	12,293	14,323	60,166	0.272		1.996	
2022		13,082	13,075	15,784	75,949	0.233		2.150	
2023		13,915	13,906	17,440	93,389	0.207		2.270	
2024		14,800	14,789	19,318	112,707	0.188		2.364	
2025		15,743	15,729	21,447	134,155	0.174		2.441	
2026		15,743	16,729	23,862	158,017	0.164		2.503	
MAPE (%)		4.33	4.40	3.89					

Table 5 ISO 14001 growth for India

Years	Actual data	EGM	DGM	NDGM	Cumulative	RGR	Mean RGR	Dt	Mean Dt
2005	1698	1698	1698	1698	1698	–			
2006	2016	2398	2405	2247	3714	0.783	0.312	0.938	2.000
2007	2640	2698	2706	2609	6354	0.537		1.315	
2008	3281	3035	3044	3003	9635	0.416		1.569	
2009	3799	3415	3425	3431	13,434	0.332		1.795	
2010	3878	3843	3853	3896	17,312	0.254		2.065	
2011	4147	4324	4335	4402	21,459	0.215		2.231	
2012	4286	4865	4878	4952	25,745	0.182		2.396	
2013	5872	5474	5488	5549	31,617	0.205		2.276	
2014	6443	6159	6174	6199	38,060	0.185		2.378	
2015	6782	6930	6946	6905	44,842	0.164		2.501	
2016	7725	7798	7815	7672	52,567	0.159		2.532	
2017		8774	8793	8506	8506				
2018		9872	9892	9413	17,919	0.745	0.309	0.987	1.998
2019		11,108	11,130	10,399	28,318	0.458		1.475	
2020		12,499	12,522	11,470	39,788	0.340		1.772	
2021		14,063	14,088	12,635	52,423	0.276		1.981	
2022		15,823	15,850	13,901	66,324	0.235		2.140	
2023		17,804	17,832	15,277	81,601	0.207		2.267	
2024		20,033	20,062	16,773	98,374	0.187		2.370	
2025		22,541	22,572	18,399	116,773	0.171		2.457	
2026		25,362	25,395	20,167	136,939	0.159		2.530	
MAPE (%)		6.52	6.30	5.88					

Similarly for China, by using the DGM time response formula based on forecasting equation parameters $\beta_1 = 1.169$ and $\beta_2 = 26{,}433.357$, Eq. (17) can be written as:

$$\hat{x}^{(1)}(N) = \left[x^{(0)}(1) - \frac{26433.357}{1-1.169}\right]1.169^N + \frac{26433.357}{1-1.169} \quad (17)$$

These β_1 and β_2 parameters can be calculated by the software mentioned in the "Methodology" section.

Moreover, for China, by NDGM forecasting equation parameters $\beta_1 = 1.128$, $\beta_2 = 2664.166$, $\beta_3 = 20{,}927.409$, and $\beta_4 = -247.952$ respectively in the time series, formula of NDGM can be obtained. These parameters can be calculated by the Matlab software. For a clear and easy comparison among actual data and NDGM simulated data from the period

Table 6 ISO 14001 certification growth for the USA

Year	Actual data	EGM	NDGM	DGM	Cumulative	RGR	Mean RGR	Dt	Mean Dt
2005	5061	5061	5061	5061	5061	–			
2006	5585	5086	5075	5095	10,646	0.744	0.246	0.989	2.373
2007	5462	5151	5164	5158	16,108	0.414		1.575	
2008	4974	5216	5245	5222	21,082	0.269		2.006	
2009	5225	5283	5317	5286	26,307	0.221		2.201	
2010	4407	5350	5383	5352	30,714	0.155		2.558	
2011	4957	5418	5442	5418	35,671	0.150		2.593	
2012	5699	5487	5496	5485	41,370	0.148		2.602	
2013	6071	5556	5544	5553	47,441	0.137		2.681	
2014	5617	5627	5588	5622	53,058	0.112		2.883	
2015	6067	5698	5627	5692	59,125	0.108		2.916	
2016	5582	5771	5663	5762	64,707	0.090		3.099	
2017		5844	5695	5833	5833				
2018		5918	5724	5906	11,739	0.699	0.262	1.051	2.214
2019		5994	5750	5979	17,718	0.412		1.581	
2020		6070	5774	6053	23,771	0.294		1.918	
2021		6147	5795	6128	29,899	0.229		2.166	
2022		6225	5815	6204	36,103	0.189		2.362	
2023		6304	5832	6281	42,383	0.160		2.523	
2024		6384	5848	6358	48,741	0.140		2.661	
2025		6465	5862	6437	55,178	0.124		2.780	
2026		6547	5875	6517	61,695	0.112		2.886	
MAPE (%)		6.65	6.83	6.42					

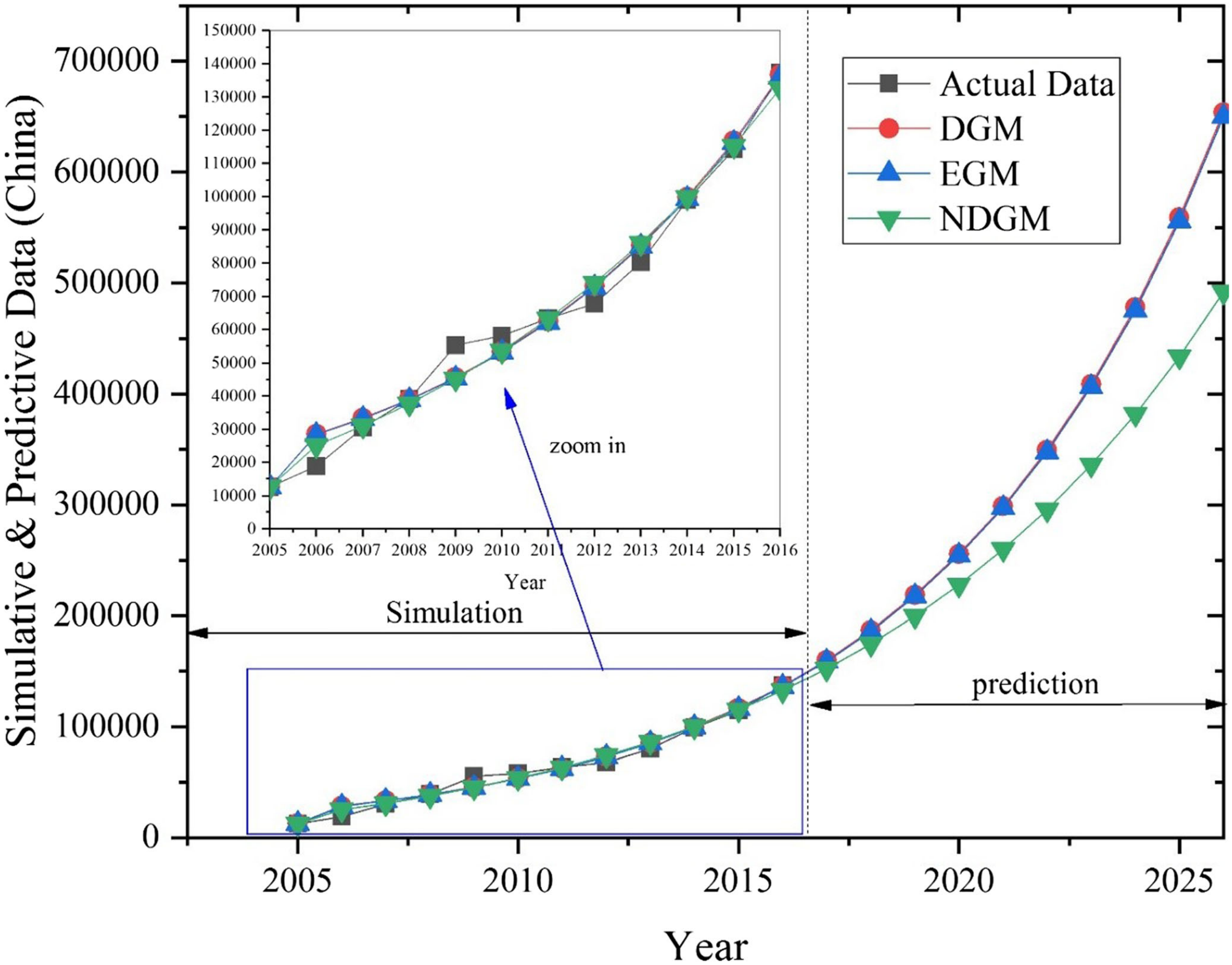

Fig. 1 Prediction of China's ISO 14001 certifications

2005–2026 against the total number of EMS, certifications have been shown in Fig. 1.

Table 2 represents Japan's results, 86.8%, 86%, and 88.10% MAPE accuracy for EGM, DGM, and NDGM respectively. Hence, NDGM shows slightly more forecasting effectiveness comparatively than EGM and DGM, as shown in Fig. 2. According to EGM parameters $a = -0.01691495$ and $b = 32{,}228.834$ Eq. (18) will be resulted:

$$x^{(0)}(N) = 1928812.087\left(1-\mathrm{e}^{-0.01691495}\right)\mathrm{e}^{0.01691495(N-1)} \quad (18)$$

Similarly for Japan, by using the DGM-based forecasting equation parameters $\beta_1 = 0.982$ and $\beta_2 = 32{,}190.501$, Eq. (19) can be presented.

$$\hat{x}^{(1)}(N) = \left[x^{(0)}(1) - \frac{32190.501}{1-0.982}\right]1.088^{N} + \frac{32190.501}{1-0.982} \quad (19)$$

Additionally, for Japan, on the basis of NDGM forecasting equation parameters $\beta_1 = 0.869$, $\beta_2 = 3459.439$, $\beta_3 = 31{,}145.425$, and $\beta_4 = -1332.496$.

Table 3 represents the results of Italy, 95.47%, 95.6%, and 96.86% MAPE accuracy for EGM, DGM, and NDGM respectively. As a result, NDGM shows slightly more forecasting effectiveness comparatively than EGM and DGM, as shown in Fig. 3. According to EGM parameters $a = -0.08488172$ and $b = 10{,}247.228$, Eq. (20) is presented:

$$x^{(0)}(N) = 127803.614\left(1-\mathrm{e}^{-0.15648621}\right)\mathrm{e}^{0.15648621(N-1)} \quad (20)$$

Similarly for Italy, by using the DGM-based forecasting equation parameters $\beta_1 = 1.088$ and $\beta_2 = 10{,}715.384$, Eq. (21) can be resulted:

$$\hat{x}^{(1)}(N) = \left[x^{(0)}(1) - \frac{10715.384}{1-1.088}\right]1.088^{N} + \frac{10715.384}{1-1.088} \quad (21)$$

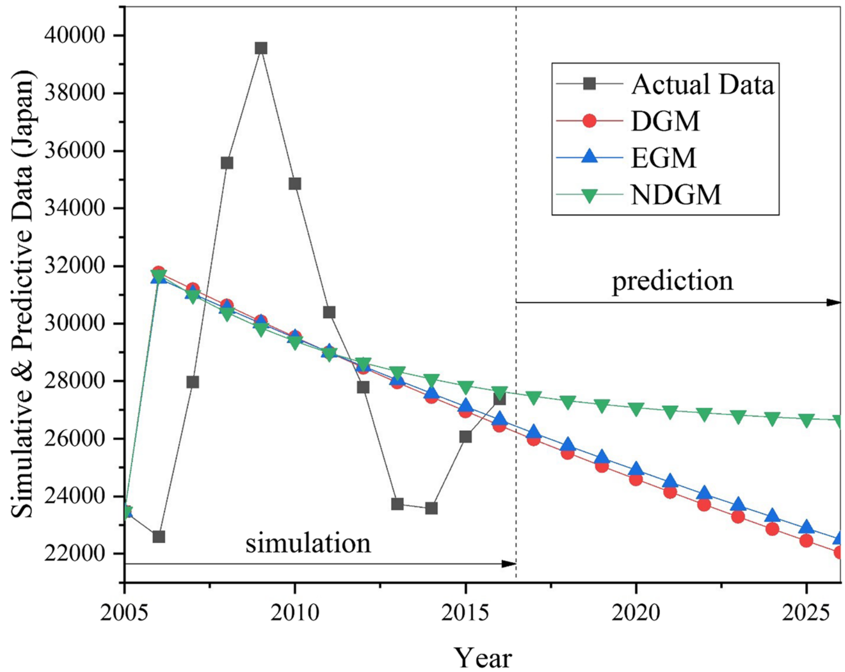

Fig. 2 Prediction of Japan's ISO 14001 certifications

Furthermore, for China, on the basis of NDGM forecasting equation parameters $\beta_1 = 0.985$, $\beta_2 = 1796.408$, $\beta_3 = 8309.663$, and $\beta_4 = 0.073$.

Table 4 represents MAPE accuracy level for Germany, 95.67%, 95.6%, and 96.11% by EGM, DGM, and NDGM respectively. Consequently, NDGM shows slightly more effectiveness comparatively than EGM and DGM, as shown in Fig. 4. According to EGM parameters $a = -0.06171916$ and $b = 4450.1471$, Eq. (22) has been resulted:

$$\hat{x}^{(0)}(N) = 7654317023\left(1 - e^{-0.15648621}\right)e^{0.15648621(N-1)} \tag{22}$$

Similarly for Germany, by using the DGM-based forecasting equation parameters $\beta_1 = 1.064$ and $\beta_2 = 4596.36$, Eq. (23) has been obtained:

$$\hat{x}^{(1)}(N) = \left[x^{(0)}(1) - \frac{4596.36}{1-1.064}\right]1.064^N + \frac{4596.36}{1-1.064} \tag{23}$$

Additionally, for Germany, by NDGM forecasting equation parameters $\beta_1 = 1.133901$, $\beta_2 = -457.165268$, $\beta_3 = 4931.3784$, and $\beta_4 = 13.453014$ respectively in the time series formula of NDGM.

Table 5 represents the results of India, 93.48%, 93.70%, and 94.12% MAPE accuracy for EGM, DGM, and NDGM respectively. Hence, NDGM shows slightly more effectiveness comparatively than EGM and DGM, as shown in Fig. 5. According to EGM parameters $a = -0.11794128$ and $b = 2058.6473$, Eq. (24) has been resulted:

$$x^0(N) = 19152.84957\left(1 - e^{-0.15648621}\right)e^{0.15648621(N-1)} \tag{24}$$

Similarly for India, by using the DGM-based forecasting equation parameters $\beta_1 = 1.125$ and $\beta_2 = 2192.669$, Eq. (25) can be obtained:

$$\hat{x}^{(1)}(N) = \left[x^{(0)}(1) - \frac{2192.669}{1-1.125}\right]1.125^N + \frac{2192.669}{1-1.125} \tag{25}$$

Moreover, for India, by NDGM forecasting equation parameters $\beta_1 = 1.087009$, $\beta_2 = 166.669348$, $\beta_3 = 1932.740253$, and $\beta_4 = 0.1909553$ respectively in the time series formula of NDGM.

Table 6 represents the results of the USA, 93.35%, 93.58%, and 93.17% MAPE accuracy for EGM, DGM, and NDGM respectively. Accordingly, DGM shows slightly more effectiveness comparatively than EGM and NDGM, as shown in Fig. 6. According to EGM parameters $a = -0.01262588$ and $b = 4990.322$, Eq. (26) has been resulted:

$$x^0(N) = 400306.4799\left(1 - e^{-0.15648621}\right)e^{0.15648621(N-1)} \tag{26}$$

Similarly for the USA, by using the DGM-based forecasting equation parameters $\beta_1 = 1.012$ and $\beta_2 = 5032.194$, Eq. (27) can be obtained:

$$\hat{x}^{(1)}(N) = \left[x^{(0)}(1) - \frac{5032.194}{1-1.012}\right]1.012^N + \frac{5032.194}{1-1.012} \tag{27}$$

Moreover, for the USA, by NDGM forecasting equation parameters $\beta_1 = 0.9030998$, $\beta_2 = 580.897527$, $\beta_3 =$

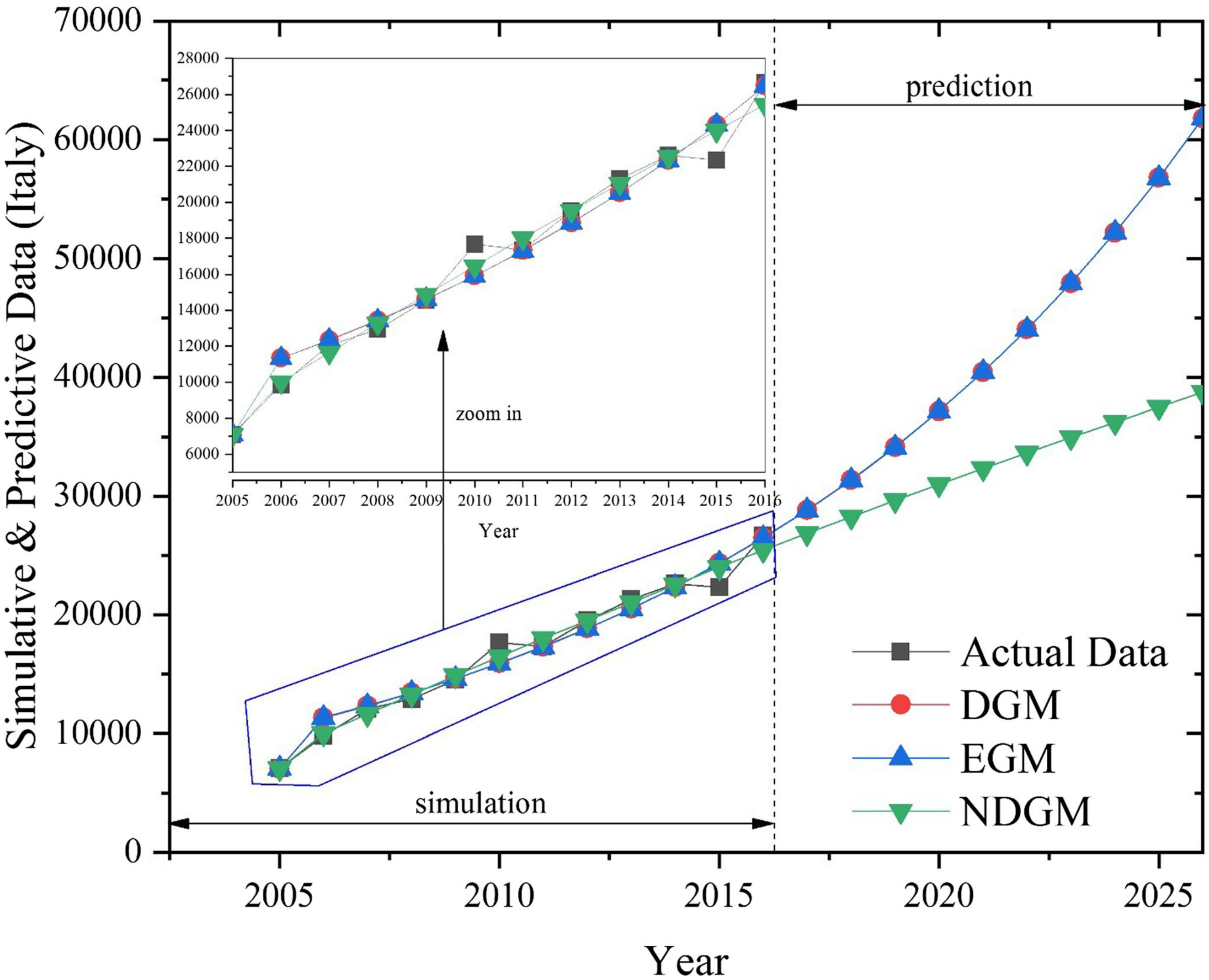

Fig. 3 Prediction of Italy's ISO 14001 certifications

4990.067165, and $\beta_4 = 56.278536$ respectively in the time series formula of NDGM.

We used three grey models to forecast ISO 14001 certification output in six countries. The results reveal that the NDGM showed more accuracy than DGM and EGM in case of China, Italy, Germany, and India, meanwhile DGM showed slightly effectiveness than EGM and NDGM in the USA case given in Table 7. According to average MAPE %, overall comparison scenario accuracy level of NDGM was 94.47%, which is slightly better than EGM 93.68% and DGM 94.04% respectively.

In contrast, wise country comparison among six countries based on relative growth rate (RGR) and doubling time (D_t) achieves interesting results, as per actual data 2005–2016 provided by ISO the RGR Eq. (28) was obtained:

$$\text{China}_{(0.374)} > \text{India}_{(0.312)} > \text{Germany}_{(0.306)} > \text{Italy}_{(0.305)} > \text{USA}_{(0.246)} > \text{Japan}_{(0.244)} \tag{28}$$

According to doubling time (D_t) comparison based on actual data between six countries, the following sequence of results was obtained:

$$\text{Japan}_{(2.381)} > \text{USA}_{(2.373)} > \text{Germany}_{(2.211)} > \text{Italy}_{(2.059)} > \text{India}_{(2.00)} > \text{China}_{(1.825)} \tag{29}$$

These two exciting findings show that the relative growth rate of developing countries (China and India) is relatively greater and for the developed countries (Germany, Italy, USA, and Japan) is relatively less great. Meanwhile, these findings also reveal that developing countries need less time to double ISO 14001 certifications numbers for a given RGR than developed countries. Therefore, relative growth of EMS in developing countries is a source of competitive benefits, more employee satisfaction, cost-saving, environmental sustainability,

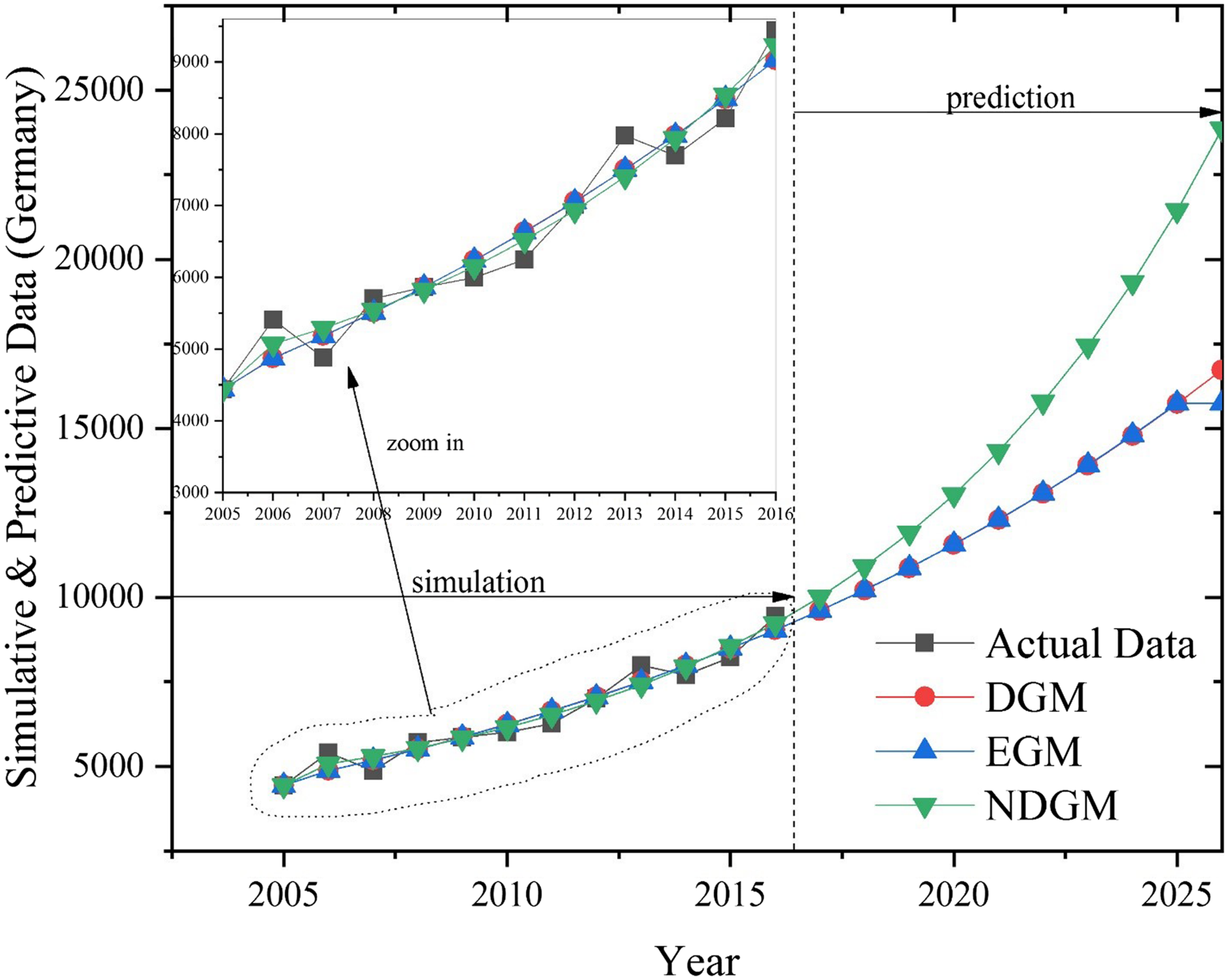

Fig. 4 Prediction of Germany's ISO 14001 certifications

improved relationship with communities, better image, and more business opportunities in the international market over strong competitors in developed countries (Hojnik and Ruzzier 2017).

However, we used NDMG-based simulated data to ascertain the situation of ISO 14001 certification in the future from 2017 to 2026. According to RGR sequence, the result was obtained in Eq. (30):

$$\begin{aligned}&\text{China}_{(0.329)} > \text{India}_{(0.309)} > \text{Germany}_{(0.307)}\\&\quad > \text{Italy}_{(0.278)} > \text{USA}_{(0.258)} > \text{Japan}_{(0.254)}\end{aligned} \tag{30}$$

The same sequence was obtained on the simulated base data; in the next 10 years, developing countries (China and India) will retain their position in terms of relative growth (3.29%) and (3.09%) respectively followed by Germany, Italy, the USA, and Japan.

According to doubling time (D_t) comparison based on simulated data between six countries, the following sequence of the result was obtained:

$$\begin{aligned}&\text{China}_{(1.916)} < \text{India}_{(1.998)} < \text{Italy}_{(2.136)}\\&\quad < \text{Germany}_{(2.000)} < \text{USA}_{(2.238)} < \text{Japan}_{(2.257)}\end{aligned} \tag{31}$$

Our finding also reveals that China needs (1.916) years to double its number of certifications because of the higher relative growth rate followed by India, Italy, Germany, the USA, and Japan.

Three different grey model results presented above by actual data and simulated base data, forecast EMS certification for the next 10 years. Now here a question arises in the long run as to who will get a maximum number of ISO 14001 accreditations. Therefore, to answer this question (Javed and Liu 2018) generated two grey synthetic indices by actual

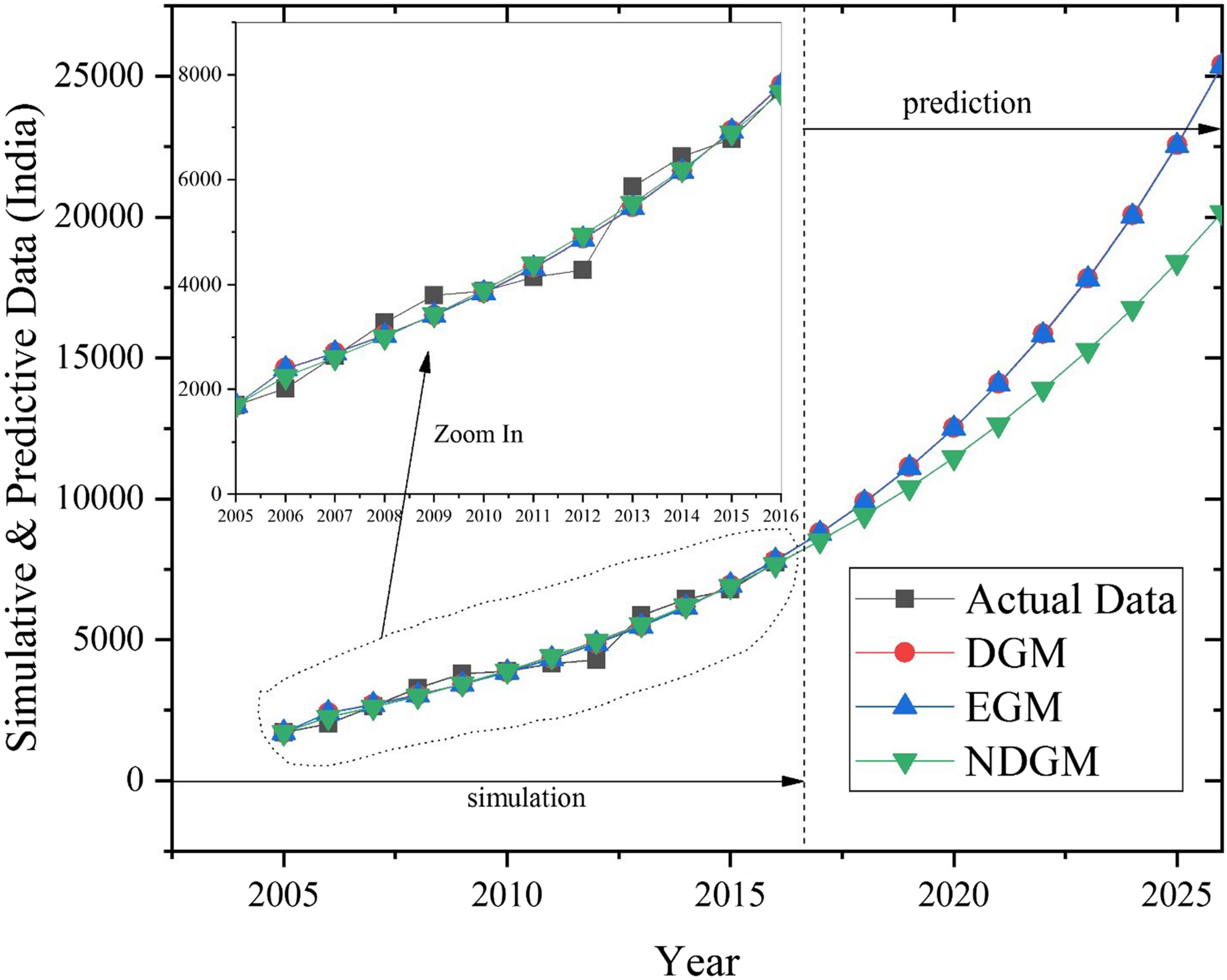

Fig. 5 Prediction of India's ISO 14001 certifications

(past) and forecasted (future) data, which implies that $RGR_{synthetic}$ indicates synthetic relative growth rate has been obtained in Eq. (32):

$$\mathrm{RGR_{synthetic}} = \theta\left(\mathrm{RGR_{past}}\right) + (1-\theta)\left(\mathrm{RGR_{future}}\right) \tag{32}$$

Where $\mathrm{RGR_{past}}$ indicates the relative growth rate got from original data and $\mathrm{RGR_{future}}$ states the Relative Growth Rate got from forecasted data. Here, θ indicates the *coefficient of relative weights* and value of θ can be taken as 0.5. If the decision makers in decision making consider the relative growth rate of original data more important, then $\theta > 1-\theta$.

Furthermore, if $D_{synthetic}$ indicates synthetic doubling time (STD) it resulted in Eq. (33):

$$\mathrm{D_{synthetic}} = \theta\left(\mathrm{D_{past}}\right) + (1-\theta)\left(D_{\mathrm{future}}\right) \tag{33}$$

Where, $\mathrm{D_{past}}$ indicates the doubling time obtained from actual data and $\mathrm{RGR_{future}}$ reports the relative growth rate got from simulated/forecasted data, and θ indicates the relative weight. Whereas, on the basis of relative growth rate by using these two grey synthetic indices, the following sequence was obtained.

$$\begin{aligned}&\mathrm{China}_{(0.338)} > \mathrm{India}_{(0.311)} > \mathrm{Germany}_{(0.307)} > \mathrm{Italy}_{(0.292)}\\&> \mathrm{USA}_{(0.252)} > \mathrm{Japan}_{(0.249)}\end{aligned} \tag{34}$$

According to doubling time (D_t) comparison based on STD between six countries, the following sequence of the result was obtained.

$$\begin{aligned}&\mathrm{Japan}_{(2.319)} > \mathrm{USA}_{(2.306)} > \mathrm{Germany}_{(2.106)}\\&> \mathrm{Italy}_{(2.089)} > \mathrm{India}_{(1.999)} > \mathrm{China}_{(1.871)}\end{aligned} \tag{35}$$

Hence, based on synthetic model feasibility, we get the sequence by applying SRGR and STD, which is similar to

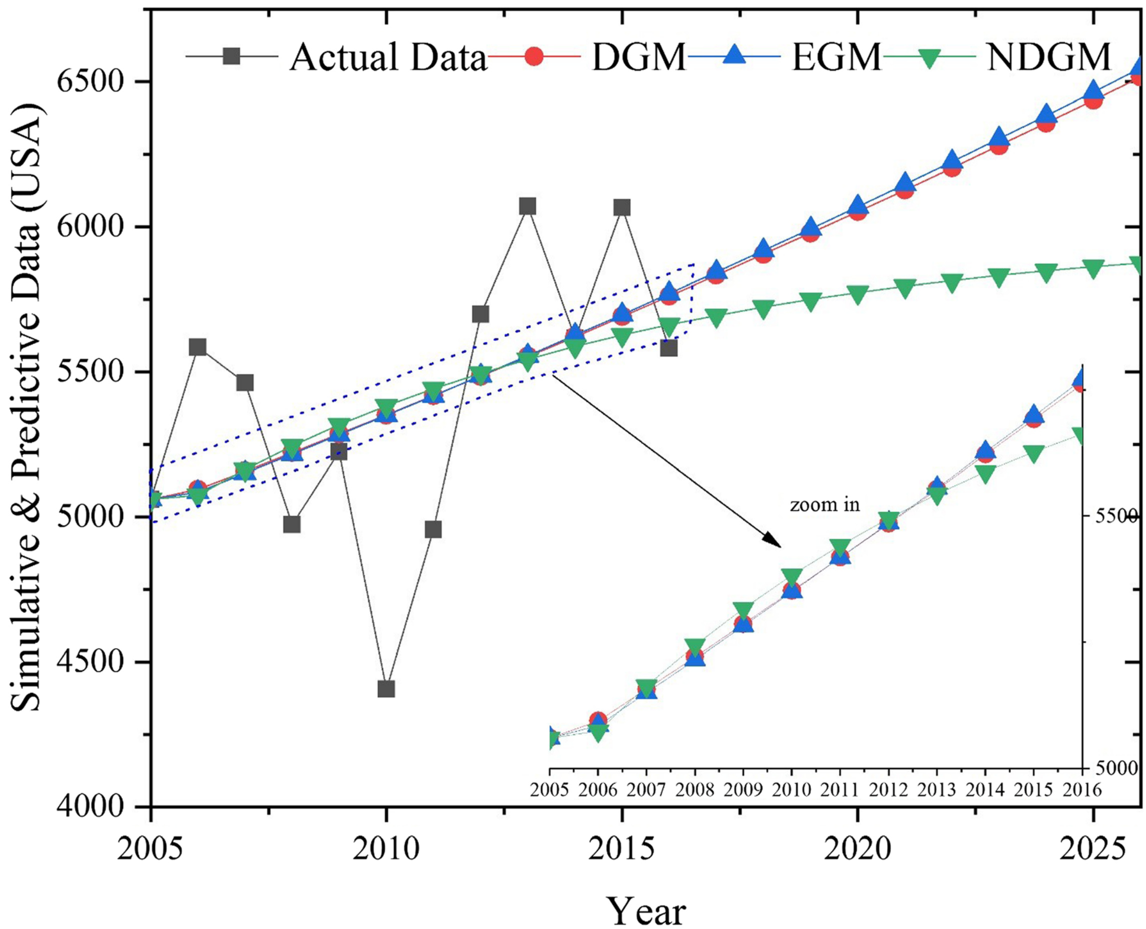

Fig. 6 Prediction of USA's ISO 14001 certifications

the sequence obtained against the actual data RGR and doubling time (D_t). Whereas, the results are aligned and confirm that ISO 14001 certification growth in developing countries is more than that of a developed country and they need less time to double in number. Our finding also reveals that in the long run China and India would have a competitive edge over other more developed countries.

Table 7 MAPE (%) comparison

	MAPE %		
	EGM	NDGM	DGM
Countries			
China	9.57	7.9	8.29
Japan	13	11.90	13.20
Italy	4.53	3.14	4.4
Germany	4.33	3.89	4.4
India	6.52	5.88	6.3
USA	6.65	6.83	6.42
Average MAPE %	6.32	5.53	5.96
Overall accuracy level %	93.68	94.47	94.04

Conclusion

We developed a forecasting framework to predict the ISO 14001 trends in China, Japan, India, the USA, Germany, and Italy by using the most popular and accurate methodology of the grey foresting system. Results presented that methodology of three grey prediction models were satisfactory and were successfully applied based on actual data to forecast the ISO 14001 trends from the period 2017 to 2026 in the USA, Japan, Germany, Italy, China, and India. All six countries show the inclined trend in ISO 14001 certification from 2015 to 2016 in Figs. 1, 2, 3, 4, 5, and 6. However, the results indicate in the long run China and India are more likely to maintain their positions as leaders in adopting the higher standard in some ISO 14001 certifications followed by Germany, Italy, the USA, and Japan. A linear trend in growth of accreditation is found in this study. The study confirms the accuracy level of NDGM is better than the other two forecasting even GM (1,1) and discrete GM (1,1) grey models. It indicates that the EMS certification growth rate in developing countries is higher whereas developed countries tend to possess less growth rate in certification and require more time to take their double certification in numbers. Furthermore, two synthetic grey models were tested successfully to analyze the relative growth rate and time expected to double the quantity of certification.

Also, the study has some policy implications for academicians and practitioners. In an academic perspective, this article provides an essential contribution to studying ISO 14001 future trends by grey forecasting models. For the practitioner community, this study can help out the International Organization for Standardization to analyze 14001 market trend until 2026. Similarly, this research can also increase the interest of certification bodies to strengthen those countries where companies' environmental sustainability requires more efforts through registration of ISO 14001. The government should take some serious steps to provide subsidies and provide training on ISO 14001 standard to make organizations more environmentally sustainable. The study can be used as a valuable awareness tool for the customers who can demand environment-friendly and sustainable products and services as a value addition in the existing product line.

Lastly, considering the challenging task of limiting global warming to 1.5 °C, the need for sustainability-oriented companies and industries is very important for the countries. The study encourages the ISO 14001 certifications for the companies who want to remain profitable as well as sustainable.

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